

CHAPTER – 15

C O N T I N U I T Y

Definition, Algebraic Continuity, Trigonometric Continuity, Exponential Continuity and Discontinuity

1. Introduction to Continuity

Basic Idea

A function is said to be **continuous** if its graph can be drawn without lifting the pen from the paper.

In simple words:

There should be no break, jump, or hole in the graph.

2. Definition of Continuity

Definition of Continuity

A function $f(x)$ is continuous at $x = a$ if:

1. $f(a)$ is defined
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

Important Formula

$$\lim_{x \rightarrow a} f(x) = f(a)$$

If this condition is true, then the function is continuous at $x = a$.

Geometric Interpretation of Continuity

Geometric Meaning of Continuity

Geometrically, a function is said to be continuous if its graph has:

- no break,
- no jump,
- no hole,
- and no sudden interruption.

In simple words:

A graph is continuous if we can draw it without lifting the pen from the paper.

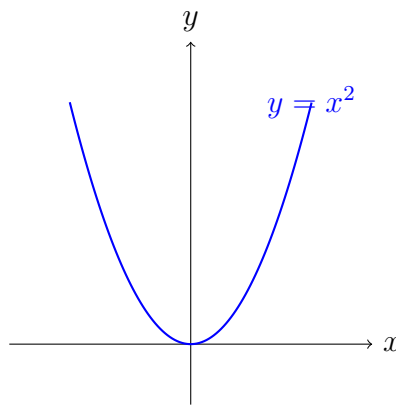
1. Continuous Function Graph

Example 1

Consider the function:

$$f(x) = x^2$$

This function is continuous for all real numbers.



The graph has no gaps or jumps.
Therefore,

$$f(x) = x^2 \text{ is continuous everywhere}$$

2. Geometric Meaning at a Point

A function is continuous at $x = a$ if:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Geometrically, this means:

- the graph approaches the same value from both sides,
- and the graph actually passes through that point.

Example 2

Consider:

$$f(x) = 2x + 1$$

at $x = 2$.

$$f(2) = 5$$

and

$$\lim_{x \rightarrow 2} (2x + 1) = 5$$

So the graph passes smoothly through the point:

$$(2, 5)$$

Hence the function is continuous at $x = 2$.

3. Geometric Interpretation of Discontinuity

If a graph contains:

- a hole,
- a jump,
- or a break,

then the function is discontinuous.

(i) Hole in Graph (Removable Discontinuity)

Example 3

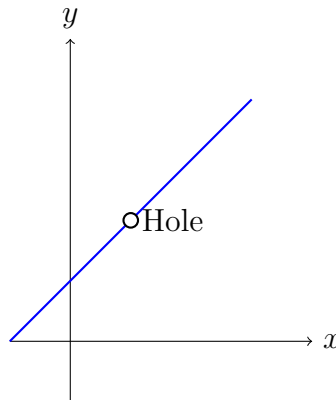
Consider:

$$f(x) = \frac{x^2 - 1}{x - 1}$$

At $x = 1$,

$$f(x) = \frac{0}{0}$$

The graph looks like a straight line with one missing point.

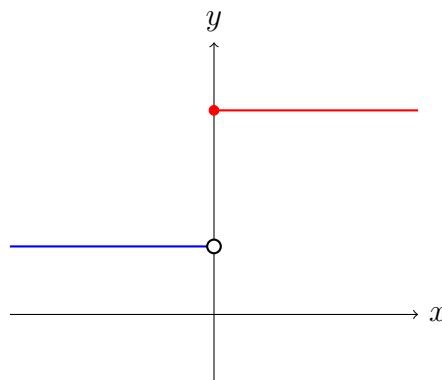


This missing point creates a discontinuity.

(ii) Jump Discontinuity

Example 4

$$f(x) = \begin{cases} 1, & x < 0 \\ 3, & x \geq 0 \end{cases}$$



The graph suddenly jumps from 1 to 3.

Therefore,

Jump discontinuity

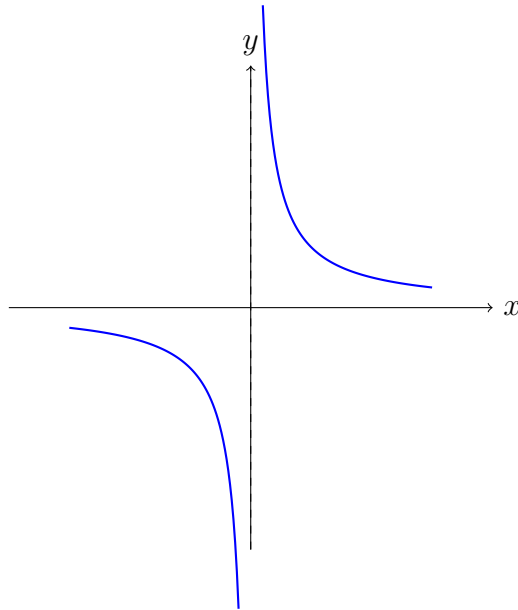
(iii) Infinite Discontinuity

Example 5

Consider:

$$f(x) = \frac{1}{x}$$

Near $x = 0$, the graph rises infinitely.



The graph becomes unbounded near $x = 0$.
Hence the function is discontinuous.

4. Important Observations

Key Points

- Continuous graphs are smooth and unbroken.
- Discontinuous graphs contain gaps or jumps.
- If the left-hand and right-hand limits are equal and equal to function value, then the graph is continuous.
- Continuity is closely related to the graphical behavior of functions.

5. Summary Table

Graph Behavior	Interpretation
Smooth curve	Continuous
Hole in graph	Removable discontinuity
Sudden jump	Jump discontinuity
Infinite rise/fall	Infinite discontinuity
Rapid oscillation	Oscillatory discontinuity

6. Practice Questions

1. Explain geometrically why $f(x) = x^2$ is continuous.
2. Draw the graph of a function with a jump discontinuity.
3. What type of discontinuity is shown by $f(x) = \frac{1}{x}$ at $x = 0$?
4. Explain the geometric meaning of:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

5. Draw a graph showing a removable discontinuity.

3. Understanding Continuity with Example

Example 1

Check continuity of

$$f(x) = 2x + 3$$

at $x = 1$.

Step 1: Find $f(1)$

$$f(1) = 2(1) + 3 = 5$$

Step 2: Find limit

$$\lim_{x \rightarrow 1} (2x + 3) = 5$$

Step 3: Compare

$$\lim_{x \rightarrow 1} f(x) = f(1)$$

Therefore,

$f(x)$ is continuous at $x = 1$

Continuity of Algebraic, Trigonometric and Exponential Functions

Introduction

Many functions used in mathematics are naturally continuous. These include:

- Algebraic functions
- Trigonometric functions
- Exponential functions

Understanding their continuity helps us study graphs, limits, derivatives, and real-life applications.

1. Continuity of Algebraic Functions

Definition

An algebraic function is formed using:

- addition,
- subtraction,
- multiplication,
- division,
- powers,
- roots

of polynomials.

Examples:

$$x^2 + 3x + 1$$

$$\frac{x + 1}{x - 2}$$

$$\sqrt{x + 4}$$

Important Theorem

Algebraic functions are continuous at every point where they are defined.

2. Continuity of Polynomial Functions

Polynomial Functions

Every polynomial function is continuous for all real numbers.

Examples:

$$x^2 + 1$$

$$3x^3 - 2x + 7$$

$$x^5 - 4x^2 + 6$$

Why are Polynomial Functions Continuous?

A polynomial contains only powers of x and constants.

Since:

$$\lim_{x \rightarrow a} x^n = a^n$$

and sums/products of continuous functions are continuous,
therefore all polynomials are continuous everywhere.

Example 1

Check continuity of

$$f(x) = x^2 + 4x + 1$$

at $x = 2$.

Step 1: Find $f(2)$

$$f(2) = 2^2 + 4(2) + 1$$

$$= 4 + 8 + 1$$

$$= 13$$

Step 2: Find limit

$$\lim_{x \rightarrow 2} (x^2 + 4x + 1) = 13$$

Step 3: Compare

$$\lim_{x \rightarrow 2} f(x) = f(2)$$

Hence,

$f(x)$ is continuous at $x = 2$

Example 2

Check continuity of

$$f(x) = 2x^3 - 5x + 1$$

at $x = -1$.

Function value

$$\begin{aligned} f(-1) &= 2(-1)^3 - 5(-1) + 1 \\ &= -2 + 5 + 1 \\ &= 4 \end{aligned}$$

Limit

$$\lim_{x \rightarrow -1} (2x^3 - 5x + 1) = 4$$

Thus,

$$\boxed{f(x) \text{ is continuous at } x = -1}$$

3. Continuity of Rational Functions

Rational Functions

A rational function is a ratio of two polynomials:

$$f(x) = \frac{p(x)}{q(x)}$$

Rational functions are continuous wherever denominator is nonzero.

Important Point

A rational function becomes discontinuous where:

$$q(x) = 0$$

because division by zero is undefined.

Example 3

Check continuity of

$$f(x) = \frac{x+3}{x-1}$$

at $x = 2$.

Function value

$$f(2) = \frac{2+3}{2-1} = 5$$

Limit

$$\lim_{x \rightarrow 2} \frac{x+3}{x-1} = 5$$

Therefore,

$$f(x) \text{ is continuous at } x = 2$$

Example 4

Check continuity of

$$f(x) = \frac{x+1}{x-4}$$

at $x = 4$.

At $x = 4$,

$$x - 4 = 0$$

Function is undefined.

Hence,

$$f(x) \text{ is discontinuous at } x = 4$$

4. Continuity of Root Functions

Root Functions

Root functions are continuous wherever the expression inside the root is defined.

Examples:

$$\sqrt{x}$$

is continuous for

$$x \geq 0$$

Example 5

Check continuity of

$$f(x) = \sqrt{x+5}$$

at $x = 4$.

$$f(4) = \sqrt{9} = 3$$

and

$$\lim_{x \rightarrow 4} \sqrt{x+5} = 3$$

Thus,

$$f(x) \text{ is continuous at } x = 4$$

5. Continuity of Trigonometric Functions

Trigonometric Functions

The trigonometric functions:

$$\sin x, \quad \cos x$$

are continuous for all real numbers.

Functions such as:

$$\tan x, \quad \sec x$$

are continuous only where they are defined.

Why are $\sin x$ and $\cos x$ Continuous?

The graphs of sine and cosine curves are smooth and unbroken.

For every real number a ,

$$\lim_{x \rightarrow a} \sin x = \sin a$$

and

$$\lim_{x \rightarrow a} \cos x = \cos a$$

Hence they are continuous everywhere.

Example 6

Check continuity of

$$f(x) = \sin x$$

at

$$x = \frac{\pi}{6}$$

Function value

$$f\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

Limit

$$\lim_{x \rightarrow \frac{\pi}{6}} \sin x = \frac{1}{2}$$

Hence,

$$\sin x \text{ is continuous at } x = \frac{\pi}{6}$$

Example 7

Check continuity of

$$f(x) = \cos x$$

at $x = 0$.

$$f(0) = 1$$

and

$$\lim_{x \rightarrow 0} \cos x = 1$$

Therefore,

$$\cos x \text{ is continuous at } x = 0$$

6. Continuity of Tangent Function

Tangent Function

$$\tan x = \frac{\sin x}{\cos x}$$

Since denominator should not be zero,

$$\tan x$$

is continuous wherever:

$$\cos x \neq 0$$

That is,

$$x \neq \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

Example 8

Check continuity of

$$f(x) = \tan x$$

at

$$x = \frac{\pi}{4}$$

$$\tan \frac{\pi}{4} = 1$$

and

$$\lim_{x \rightarrow \frac{\pi}{4}} \tan x = 1$$

Hence,

$\tan x$ is continuous at $x = \frac{\pi}{4}$

Example 9

Check continuity of

$$f(x) = \tan x$$

at

$$x = \frac{\pi}{2}$$

At

$$x = \frac{\pi}{2}$$

$$\cos \frac{\pi}{2} = 0$$

Thus tangent is undefined.

Therefore,

$\tan x$ is discontinuous at $x = \frac{\pi}{2}$
--

7. Continuity of Exponential Functions

Exponential Functions

Exponential functions are continuous for all real numbers.

Examples:

$$e^x, \quad 2^x, \quad 10^x$$

Why are Exponential Functions Continuous?

The graph of an exponential function is smooth and never breaks.

For every real number a ,

$$\lim_{x \rightarrow a} a^x = a^a$$

Hence exponential functions are continuous everywhere.

Example 10

Check continuity of

$$f(x) = 2^x$$

at $x = 3$.

$$f(3) = 2^3 = 8$$

and

$$\lim_{x \rightarrow 3} 2^x = 8$$

Thus,

$$\boxed{2^x \text{ is continuous at } x = 3}$$

Example 11

Check continuity of

$$f(x) = e^x$$

at $x = 0$.

$$f(0) = 1$$

and

$$\lim_{x \rightarrow 0} e^x = 1$$

Hence,

$$e^x \text{ is continuous at } x = 0$$

8. Combined Continuous Functions

Combination Theorem

If two functions are continuous, then:

- Sum is continuous

$$f(x) + g(x)$$

- Difference is continuous

$$f(x) - g(x)$$

- Product is continuous

$$f(x)g(x)$$

- Quotient is continuous if denominator is nonzero

$$\frac{f(x)}{g(x)}$$

Example 12

Check continuity of

$$f(x) = x^2 + \sin x$$

at $x = 1$.

Both x^2 and $\sin x$ are continuous.

Therefore their sum is continuous.

Hence,

$$x^2 + \sin x \text{ is continuous at } x = 1$$

9. Summary Table

Function Type	Continuity
Polynomial Functions	Continuous everywhere
Rational Functions	Continuous where denominator $\neq 0$
Root Functions	Continuous where defined
$\sin x, \cos x$	Continuous everywhere
$\tan x, \sec x$	Continuous where defined
Exponential Functions	Continuous everywhere

10. Practice Questions

1. Check continuity of

$$f(x) = x^3 + 2x - 5$$

at $x = 1$.

2. Determine whether

$$f(x) = \frac{x+2}{x-3}$$

is continuous at $x = 3$.

3. Check continuity of

$$f(x) = \sqrt{x+9}$$

at $x = 0$.

4. Determine continuity of

$$f(x) = \sin x$$

at $x = \pi$.

5. Check continuity of

$$f(x) = \tan x$$

at

$$x = \frac{\pi}{2}$$

6. Check continuity of

$$f(x) = 5^x$$

at $x = 2$.

7. Determine continuity of

$$f(x) = x^2 + \cos x$$

at $x = 0$.

8. Check continuity of

$$f(x) = \frac{\sin x}{x + 1}$$

at $x = 0$.

11. Answer Key

1. Continuous
2. Discontinuous
3. Continuous
4. Continuous
5. Discontinuous
6. Continuous
7. Continuous
8. Continuous

Types of Discontinuities

Introduction

A function is said to be **discontinuous** at a point if it fails to satisfy the conditions of continuity.

That means at least one of the following is not true:

1. $f(a)$ is defined
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

Discontinuities are classified into different types depending on the behavior of the graph near that point.

1. Removable Discontinuity

Definition

A removable discontinuity occurs when the limit exists, but the function is either:

- not defined at that point, or
- defined with a different value.

It is called removable because we can remove the discontinuity by redefining the function properly.

Geometric Meaning

Geometrically, removable discontinuity appears as a **hole** in the graph.

Example 1

Consider:

$$f(x) = \frac{x^2 - 4}{x - 2}$$

Step 1: Substitute $x = 2$

$$= \frac{0}{0}$$

The function is undefined at $x = 2$.

Step 2: Factorize

$$x^2 - 4 = (x - 2)(x + 2)$$

$$f(x) = x + 2$$

for $x \neq 2$.

Step 3: Find limit

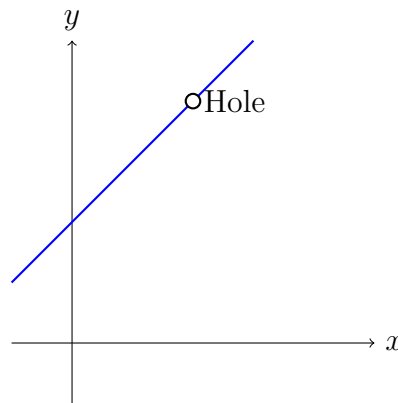
$$\lim_{x \rightarrow 2} f(x) = 4$$

The limit exists, but function is undefined at $x = 2$.

Hence,

Removable discontinuity at $x = 2$

Graphical Interpretation



2. Jump Discontinuity

Definition

A jump discontinuity occurs when:

$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$$

The left-hand and right-hand limits exist but are different.

Geometric Meaning

The graph suddenly jumps from one value to another.

Example 2

Consider:

$$f(x) = \begin{cases} 2, & x < 1 \\ 5, & x \geq 1 \end{cases}$$

Left-Hand Limit

$$\lim_{x \rightarrow 1^-} f(x) = 2$$

Right-Hand Limit

$$\lim_{x \rightarrow 1^+} f(x) = 5$$

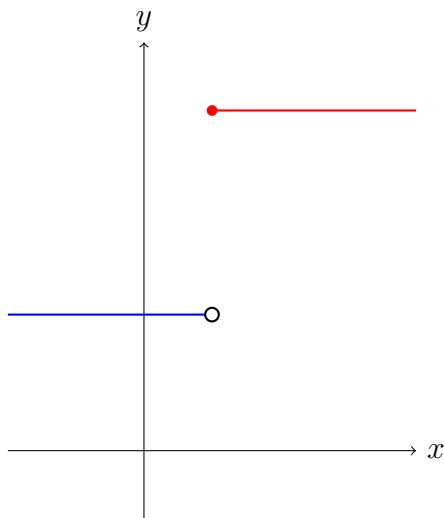
Since

$$2 \neq 5$$

therefore,

Jump discontinuity at $x = 1$

Graphical Interpretation



3. Infinite Discontinuity

Definition

An infinite discontinuity occurs when the function becomes infinitely large near a point. Usually,

$$\lim_{x \rightarrow a} f(x) = \infty$$

or

$$\lim_{x \rightarrow a} f(x) = -\infty$$

Geometric Meaning

The graph rises or falls endlessly near the discontinuity.
Such points usually create vertical asymptotes.

Example 3

Consider:

$$f(x) = \frac{1}{x-3}$$

At $x = 3$,
denominator becomes zero.

Right-Hand Limit

$$\lim_{x \rightarrow 3^+} \frac{1}{x-3} = \infty$$

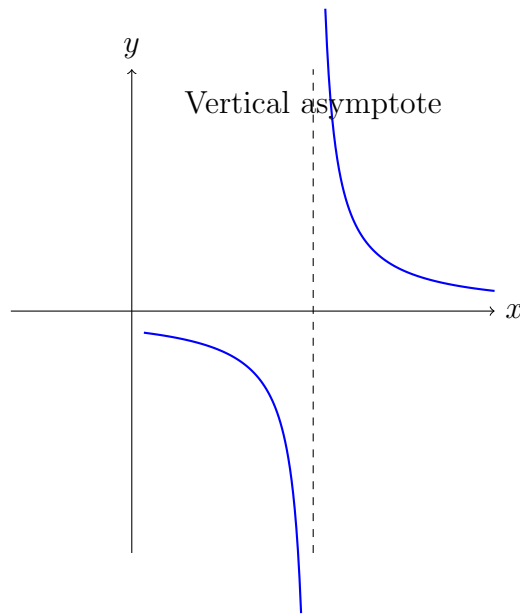
Left-Hand Limit

$$\lim_{x \rightarrow 3^-} \frac{1}{x-3} = -\infty$$

Hence,

Infinite discontinuity at $x = 3$

Graphical Interpretation



4. Oscillatory Discontinuity

Definition

An oscillatory discontinuity occurs when the function oscillates infinitely near a point and does not approach a single value.

Geometric Meaning

The graph fluctuates rapidly and never settles to one value.

Example 4

Consider:

$$f(x) = \sin \frac{1}{x}$$

as $x \rightarrow 0$.

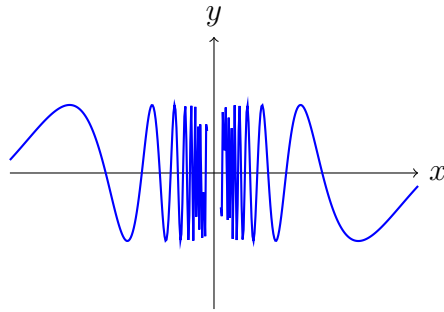
Near $x = 0$,

$$\frac{1}{x}$$

becomes very large, causing sine function to oscillate infinitely.
Thus the limit does not exist.

Oscillatory discontinuity at $x = 0$

Graphical Interpretation



5. Summary Table

Type	Main Feature	Graph Appearance
Removable	Limit exists but function improper	Hole
Jump	$LHL \neq RHL$	Sudden jump
Infinite	Function tends to $\pm\infty$	Vertical asymptote
Oscillatory	Rapid oscillation	Wavy oscillation

6. Comparison of Different Discontinuities

Type	Limit Exists?	Can be Removed?
Removable	Yes	Yes
Jump	No	No
Infinite	No	No
Oscillatory	No	No

7. Practice Questions

- Identify the type of discontinuity:

$$f(x) = \frac{x^2 - 9}{x - 3}$$

at $x = 3$.

2. Determine the discontinuity type:

$$f(x) = \begin{cases} 1, & x < 0 \\ 4, & x \geq 0 \end{cases}$$

at $x = 0$.

3. Find the discontinuity type of:

$$f(x) = \frac{1}{x+2}$$

at $x = -2$.

4. Determine the type of discontinuity:

$$f(x) = \sin \frac{1}{x}$$

at $x = 0$.

5. Explain why removable discontinuity is called removable.

Important Note

To identify the type of discontinuity:

- Check whether the limit exists.
- Compare left-hand and right-hand limits.
- Observe whether the function becomes infinite.
- Study the graphical behavior near the point.

10. Solved Examples

Example 12

Check continuity of

$$f(x) = x^2 - 4x + 1$$

at $x = 3$.

Solution

$$f(3) = 9 - 12 + 1 = -2$$

and

$$\lim_{x \rightarrow 3} (x^2 - 4x + 1) = -2$$

Therefore,

$$f(x) \text{ is continuous at } x = 3$$

Example 13

Check continuity of

$$f(x) = \frac{x+2}{x-5}$$

at $x = 5$.

Solution

Denominator becomes zero.

Hence function is undefined at $x = 5$.

$$f(x) \text{ is discontinuous at } x = 5$$

Example 14

Check continuity of

$$f(x) = e^x$$

at $x = 0$.

Solution

$$f(0) = 1$$

and

$$\lim_{x \rightarrow 0} e^x = 1$$

Therefore,

$$e^x \text{ is continuous at } x = 0$$

Example 15

Check continuity of

$$f(x) = |x|$$

at $x = 0$.

Solution

LHL:

$$0$$

RHL:

$$0$$

and

$$f(0) = 0$$

Thus,

$$\boxed{|x| \text{ is continuous at } x = 0}$$

11. Exercise

1. Check continuity of

$$f(x) = 3x^2 + 2x - 1$$

at $x = 2$.

2. Determine whether

$$f(x) = \frac{x+1}{x-4}$$

is continuous at $x = 4$.

3. Check continuity of

$$f(x) = \sin x$$

at $x = \pi$.

4. Check continuity of

$$f(x) = 3^x$$

at $x = 1$.

5. Determine continuity of

$$f(x) = |x - 2|$$

at $x = 2$.

6. Find discontinuity type:

$$f(x) = \frac{1}{x-1}$$

at $x = 1$.

7. Determine continuity of

$$f(x) = \begin{cases} 2x + 1, & x < 1 \\ 5, & x = 1 \\ x + 3, & x > 1 \end{cases}$$

at $x = 1$.

8. Find value of a for continuity:

$$f(x) = \begin{cases} ax + 1, & x < 2 \\ 7, & x = 2 \\ x + a, & x > 2 \end{cases}$$

9. Determine type of discontinuity:

$$f(x) = \sin \frac{1}{x}$$

at $x = 0$.

10. Check continuity of

$$f(x) = \tan x$$

at $x = \frac{\pi}{6}$.

12. Answer Key

1. Continuous
2. Discontinuous
3. Continuous
4. Continuous
5. Continuous
6. Infinite discontinuity
7. Discontinuous

8. $a = 3$
9. Oscillatory discontinuity
10. Continuous

Advanced Solved Examples on Continuity

Example 1

Check continuity of

$$f(x) = \begin{cases} x^2 + 1, & x < 2 \\ 5, & x = 2 \\ 3x - 1, & x > 2 \end{cases}$$

at $x = 2$.

Step 1: Find Left-Hand Limit

$$\lim_{x \rightarrow 2^-} f(x) = 2^2 + 1 = 5$$

Step 2: Find Right-Hand Limit

$$\lim_{x \rightarrow 2^+} f(x) = 3(2) - 1 = 5$$

Step 3: Find Function Value

$$f(2) = 5$$

Since

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$$

therefore,

$f(x) \text{ is continuous at } x = 2$

Example 2

Find value of k so that

$$f(x) = \begin{cases} kx + 1, & x < 3 \\ 7, & x = 3 \\ 2x + k, & x > 3 \end{cases}$$

is continuous at $x = 3$.

Step 1: Use continuity condition

For continuity:

$$\lim_{x \rightarrow 3} f(x) = f(3) = 7$$

Step 2: Left-Hand Limit

$$3k + 1 = 7$$

$$3k = 6$$

$$k = 2$$

Step 3: Right-Hand Limit

$$2(3) + k = 7$$

$$6 + k = 7$$

$$k = 1$$

Since k cannot have two values simultaneously,

No value of k makes function continuous

Example 3

Check continuity of

$$f(x) = \frac{x^2 - 9}{x - 3}$$

at $x = 3$.

Step 1: Direct substitution

$$= \frac{0}{0}$$

Function is undefined.

Step 2: Factorize

$$x^2 - 9 = (x - 3)(x + 3)$$

$$f(x) = x + 3$$

for $x \neq 3$.

Step 3: Find limit

$$\lim_{x \rightarrow 3} f(x) = 6$$

Since $f(3)$ is undefined,

$$f(x) \text{ is discontinuous at } x = 3$$

This is a removable discontinuity.

Example 4

Check continuity of

$$f(x) = \tan x$$

at

$$x = \frac{\pi}{2}$$

$$\tan x = \frac{\sin x}{\cos x}$$

At

$$x = \frac{\pi}{2}$$

$$\cos \frac{\pi}{2} = 0$$

Thus function becomes undefined.

Hence,

$$\tan x \text{ is discontinuous at } x = \frac{\pi}{2}$$

Example 5

Check continuity of

$$f(x) = \begin{cases} \sin x, & x < 0 \\ x + 1, & x \geq 0 \end{cases}$$

at $x = 0$.

Left-Hand Limit

$$\lim_{x \rightarrow 0^-} \sin x = 0$$

Right-Hand Limit

$$\lim_{x \rightarrow 0^+} (x + 1) = 1$$

Since

$$0 \neq 1$$

therefore,

$$\boxed{f(x) \text{ is discontinuous at } x = 0}$$

Example 6

Check continuity of

$$f(x) = \frac{\sqrt{x+4} - 2}{x}$$

at $x = 0$.

Step 1: Substitute

$$= \frac{0}{0}$$

Step 2: Rationalize

$$\begin{aligned} &= \frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} \\ &= \frac{x}{x(\sqrt{x+4} + 2)} \\ &= \frac{1}{\sqrt{x+4} + 2} \end{aligned}$$

Step 3: Evaluate limit

$$\lim_{x \rightarrow 0} f(x) = \frac{1}{4}$$

But function is undefined at $x = 0$.

Therefore,

$$\boxed{f(x) \text{ is discontinuous at } x = 0}$$

Example 7

Determine continuity of

$$f(x) = |x - 2|$$

at $x = 2$.

Left-Hand Limit

For $x < 2$,

$$|x - 2| = 2 - x$$

$$\lim_{x \rightarrow 2^-} |x - 2| = 0$$

Right-Hand Limit

For $x > 2$,

$$|x - 2| = x - 2$$

$$\lim_{x \rightarrow 2^+} |x - 2| = 0$$

Function value

$$f(2) = 0$$

Hence,

$ x - 2 $ is continuous at $x = 2$

Example 8

Find value of a such that

$$f(x) = \begin{cases} ax + 3, & x < 1 \\ 5, & x = 1 \\ x^2 + a, & x > 1 \end{cases}$$

is continuous at $x = 1$.

For continuity,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

Left-Hand Limit

$$a(1) + 3 = 5$$

$$a = 2$$

Right-Hand Limit

$$1 + a = 5$$

$$a = 4$$

Since values are different,

No value of a exists

Example 9

Check continuity of

$$f(x) = \frac{\sin x}{x}$$

at $x = 0$, if

$$f(0) = 1$$

Using standard limit:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

and

$$f(0) = 1$$

Thus,

$$\lim_{x \rightarrow 0} f(x) = f(0)$$

Hence,

$f(x)$ is continuous at $x = 0$

Example 10

Check continuity of

$$f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x \neq 1 \\ 3, & x = 1 \end{cases}$$

at $x = 1$.

Step 1: Simplify

$$x^2 - 1 = (x - 1)(x + 1)$$

Thus,

$$f(x) = x + 1$$

for $x \neq 1$.

Step 2: Find limit

$$\lim_{x \rightarrow 1} f(x) = 2$$

Step 3: Compare with function value

$$f(1) = 3$$

Since

$$2 \neq 3$$

therefore,

$f(x) \text{ is discontinuous at } x = 1$

This is a removable discontinuity.

Important Observation

For advanced continuity problems:

- Always check:

$$\lim_{x \rightarrow a^-} f(x), \quad \lim_{x \rightarrow a^+} f(x), \quad f(a)$$

- Piecewise functions usually require separate left and right limits.
- Rational functions become discontinuous where denominator is zero.
- Trigonometric continuity depends on whether the function is defined at that point.
- To remove removable discontinuity, redefine the function properly.